

A fuzzy convolutional neural network for enhancing multi-focus image fusion[☆]

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ARTICLE INFO

Keywords:

Deep learning
Fusion
Fuzzy sets
Multi-focus images

ABSTRACT

The images captured by the cameras contain distortions, misclassified pixels, uncertainties and poor contrast. Therefore, the multi-focus image fusion (MFIF) integrates various input image features to produce a single fused image using all its objects in focus. However, it is computationally complex, which leads to inconsistency. Hence, the MFIF method is employed to generate the fused image by integrating the fuzzy sets (FS) and convolutional neural network (CNN) to detect focused and unfocused parts in both source images. It is also compared with other competing six MFIF methods like Neutrosophic set based stationary wavelet transform (NSWT), guided filters, CNN, ensemble CNN, image fusion-based CNN and deep regression pair learning (DRPL). Benchmark datasets validate the superiority of the proposed FCNN method in terms of four non-reference assessment measures having mutual information (1.1678), edge information (0.7281), structural similarity (0.9850) and human perception (0.8020) and two reference metrics such as Peak signal-to-noise ratio (57.23) and root mean square error (1.814).

1. Introduction

Digital composite photographs can capture images with the help of photographic lenses such as digital single-lens reflex cameras. Sometimes, certain parts of the image are clear and focused, while others are unfocused because lenses have limited depth of focus. Therefore, the subject under a particular focal setting appears sharp while the rest seems blurred. To focus all the parts of an image, one needs to take pictures with a considerable depth of focus [1,2]. Therefore, the image fusion [3] approach is adopted to integrate the input images for better visibility and quality with the entire focus.

Multi focus image fusion (MFIF) is a renowned approach that captures all-in-focused images taken from the same scene having different focal settings [4]. It is also valuable for other fields like multi-modal [5] and visible infrared image fusion [6,7] as multi-focus images are highly used [8]. Presently, numerous MFIF methods exist that can be categorized as spatial domain and transform domain. However, researchers

mostly prefer to transform domain methods [9] due to their intuitive strategy to solve the task of fusion. It is a three-step approach:

1. Convert an image to the transform domain.
2. Then, fusion rules are utilized for the integration of the decomposed coefficients to generate composite coefficients.
3. Lastly, convert these coefficients to a spatial domain for producing a final fused image.

Some instances of transform domains include pyramid-based approaches such as Laplacian and morphological pyramids. More transform domain approaches include wavelet transform, gradient, sparse representation, discrete cosine transform [10] and neutrosophic set and stationary wavelet transform (NSWT) [11]. The critical issues of these methods are their total dependency on human-based transformations for the localization of reliable features. Moreover, it used hand-crafted features to represent images which ultimately degraded the image

[☆] This paper has been recommended for acceptance by Petia Radeva.

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<https://doi.org/10.1016/j.jvcir.2022.103485>

Received 20 November 2020; Received in revised form 7 December 2021; Accepted 2 March 2022

Available online 9 March 2022

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quality. In addition, three-step decomposition led to crucial information loss, hence, produced degenerated images. Also, the time and cost-intensive nature is the main obstacle to being deployed for real-time applications.

Researchers put enormous efforts into spatial domain methods where images cannot convert to other representations to solve the above problems. These methods are block or region-based and pixel-based approaches. Region-based approaches [12] such as boosted random walk (BRW) [13] and adaptive content blurring (CAB) [14] partitioned the input images into regions or blocks based on some rules and identified the focused areas such as gradients, variance and spatial details. Then, these focused regions are combined to create an enhanced single focused image. This process generated blurriness. However, composite images are processed by averaging the pixel coefficients in pixel-based approaches such as guided filters [15]. Although the procedure is simple and less time-intensive, the final image suffers from ghosting artifacts.

In recent years, fuzzy sets have been attracted to the researcher's attention and successfully applied in image processing [16,17]. During the acquisition of images, unavoidable noise, impreciseness and uncertainty are present due to which accurate visual effects of images are impossible. Therefore, to achieve the purpose of human eye observation, fuzzy sets have been adopted by the researchers. Dai et al. [18] introduced the fuzzy feature transfer-based infrared, visible fusion method, whereas Zhang et al. [19] integrated the infrared and visible images using intuitionistic fuzzy sets. First, intuitionistic fuzzy sets were employed to solve the issue of fuzziness and impreciseness. Then, Martino et al. [20] designed fuzzy transform-based image fusion that handled the errors induced in the block division of the image that occurred throughout the compression phase.

Further, Kangjian et al. [21] employed the fuzzy sets to represent the vague features presented in the low-pass sub-images in the non-subsampled contourlet transform domain. Unfortunately, this method attained poor mutual information (MI) value than the convolutional neural network (CNN) process in terms of quantitative results. For other metrics, obtained values were just comparable. Moreover, it is widely known that the fusion rules and feature extraction (activity level measurement) are the key factors to be determined both in spatial and transform domain methods. The existing one [22] focused on the high frequency features extraction by designing the filters to select the fused images using different fusion rules. But this is a manual task resulting in a poor-quality picture.

The prior study developed numerous methods by manually designing all fusion rules but feature extraction is time-consuming and requires more human efforts and is less efficient than automatically generated methods. Fortunately, recently deep learning techniques are thriving in numerous computer vision applications. Therefore, the relevance of deep learning, especially CNN is gaining more attention in MFIF fusion and de-focus blur detection. Due to its powerful capability of self-learning and avoiding designing hand-crafted features that are less prone to errors.

Defocus blur detection is the process of separation of blurred and non-blurred parts from a single image. Presently, defocus blur detection to obtain a defocus map is utilized by numerous researchers. In this direction, Zhao et al. [23] introduce a multi-stream bottom-top-bottom fully convolutional network. First, images were handled in multi-scale and then the fusion and reconstruction strategy was applied to obtain the blur maps. Then, this study refined the obtained results by the design of CRLNet [24]. However, they have added 1100 images in their dataset to evaluate the performance of the models further. Even though of impressive results, high computation cost hinders it from being used in real-time applications. Afterwards, Cun et al. [25] instilled the depth information in defocus blur detection whereas distilled the network with the knowledge of the pre-trained depth model. And their model outperformed the other 11 benchmark methods in terms of accuracy. Subsequently, Zhang et al. [26] overcame the problem of low processing

speed by developing a real-time light modified U-Net models. At the same time, Tang et al. [27] devised a neural network architecture, namely DeFusionNet that recurrently fused and advanced multi-scale deep features.

A significant limitation of these models is the requirement of massive pixel-level annotated labels. It is costly as well as computationally expensive. Therefore, Zhao et al. [28] tackled this issue by developing a weakly supervised recurrent constraint network for the focus region detection by employing box-level supervision instead of pixel-level annotations, accelerating their training speed. Moreover, they have constructed a dataset; namely, FocusBox consisting of 5000 images, which added advantage to their model.

Different from the above-discussed methods, Zhao et al. [29] explored the ensemble learning network to boost the diversity of defocus blur detectors. This network generates both high-and low-level semantic information. Finally, the ensemble network cancels out each other error detection results. Then, Li et al. [30] designed a dual branch architecture called layer-output guided complementary attention network (Local). Here, information is obtained from each unfocused image. Although it is a time-efficient approach, it is still necessary to include more complex and challenging blur image scenes for different applications.

However, Liu et al. [22] were the first authors who introduced the CNN in the domain of MFIF which produced good results. The critical drawback was using only referenced metrics to validate their approach, whose results were not satisfactory. After that, Du et al. [31] introduced the deep support value of CNN architecture. They employed the structural risk minimization loss function to train CNN. And max-pooling layers were replaced by CNN layers to avoid information loss. Finally, Lai et al. [32] presented the multi-scale visual attention deep CNN (MADCNN) approach. They developed two different units to achieve the MFIF. Multi-scale features were extracted and combined for the extraction of spatial information by the first unit. Afterwards, by the second unit, identification of the focused regions were accomplished by removing unwanted elements. The limitation was the high computational load which further made the model unsuitable for real-time applications.

Amin-Naji et al. [33] designed an ensemble CNN (ECNN) using three CNNs. The training of CNN used the three datasets to obtain the initial segmented decision maps. Although obtained fused images required no post-processing; however, the model evaluation uses only reference metrics, significantly limiting its usefulness. Wang et al. [34] integrated the deep convolutional networks and patch-based reconstruction approaches to achieve MFIF. The drawback was its incapability to fuse the areas of the images with overlapped background and foreground. Moreover, the post-processing strategy has been applied to refine the decision maps.

Gai et al. [35] designed a complex MFIF method based on two stages of CNN. First, DenseNet generated the fused image with a blurring artifact. Then, edge-deblurring generative adversarial networks (EDGAN) were employed to remove the blocking effect. However, the model attained better results with the representative approaches. The complexity of the model made the fusion process more complex and cumbersome to achieve. Finally, Zhang et al. [36] presented the transform domain image fusion-based CNN known as IFCNN. However, the developed model fused the different images such as multi-focus, visible-infrared, multi-modal and multi-exposure images. Hence, the comparative performance analysis with the existing methods indicated that the performance of their model needs to be improved to obtain a better-quality image.

Li et al. [37] propounded the MFIF method called deep regression pair learning (DRPL). In this step, the pixel-to-pixel regression strategy determined the boundary regions by directly mapping an image to a mask. Consequently, a pair learning technique obtained two binary maps. Lastly, the structural similarity index preserved the textures and edges of the final fused images. The vital issue of this model was its incapability to align multi-focus images. Existing methods attained

Table 1

Comparison of MFIF methods performance assessment for multi-focus images.

Year	Method	No of images	Metrics	Value
2017	CNN [22]	40 lytro dataset images	MI	1.14
			EI	0.74
			SS	0.98
			HP	0.79
2019	DVCNN [31]	Multi-focus images	MI	18.58
			EI	1.49
			SS	0.9806
			MI	8.6892
2019	CNN based on ensemble learning [33]	28 set of multi-focus images300 images from COCO 2014 (training)	EI	0.8597
			SS	0.9850
			VIF	0.7572
			EI	0.74
2019	CAB [14]	34 sets of both gray and colored images	MI	1.2
			FMI	0.67
			SS	0.98
			CC	0.84
2019	GF [15]	37 sets of both gray and colored images	EI	0.71
			PC	0.83
			SS	0.98
			HP	0.79
2020	IFCNN [36]	20 set of lytro multi-focus datasets 100,000 pairs NYU-D2 dataset (training)	FMI	0.615
			MI	0.8535
			SS	0.632
			SF	19.01
2020	DRPL [37]	26 pair of lytro multi-focus datasets 100 synthetic images 20,000 sets of images from the large-scale ImageNet (training)	AG	2.844
			VIF	0.9821
			For 100 synthetic dataset:	44.05
			For real dataset:	0.99
2021	NSWT [11]	30 sets of both gray images	PSNR	dataset:
			SS	8.2
			For 5 pair of real dataset:	0.75
			MI	6.8
2021	NSWT [11]	30 sets of both gray images	SF	70.9
			AG	0.94
			EI	
			VIF	
2021	NSWT [11]	30 sets of both gray images	SD	33.95
			CC	3.87
			SNR	2.40
			RMSE	43.36
2021	NSWT [11]	30 sets of both gray images	PFE	1.45
			Mean	0.84
			PSNR	0.97
			MI	0.99
2021	NSWT [11]	30 sets of both gray images	UQI	125.21
			SS	60.40

COCO: common objects in context; RMSE: root mean square error; VIF: visual information fidelity; VOC: visual object classes; P-CNN: pixel-wise CNN; PC: phase congruency; PSNR: peak signal-to-noise ratio; PFE: percentage fit error; AG: average gradient; SF: spatial frequency; SD: standard deviation; EI: edge information; FMI: feature mutual information; SNR: signal to noise ratio; SAM: spectral angle mapper; SCC: spatial correlation coefficient; MI: mutual information. SS: structural similarity; Q_w : correlation loss; UQI: universal quality index.

better metric value for some images limited their model's performance. Hence, the model needed to be optimized in terms of quantitative results. Wang et al. [38] designed a generative adversarial network named MFIF-GAN to fuse multi-focus images. Defocus spread effect (DSE) attenuated with the generation of focus maps where the foreground region was larger than corresponding objects. Model ability to simulate the DSE subsequently alleviated the detrimental effects. The downfall of this model was its inability to extract features from the gray images. Also, focus maps required post-processing. Therefore, a model needs to be optimized to alleviate the performance degradation.

Based on the analysis, MFIF is not easy to obtain. There are many vital challenges such as: (1) to efficaciously evaluate the quality of pixels by extracting focused and unfocused features to fuse them to generate an

entirely focused image is a difficult task. (2) Analyzation and quantification of the quality of the pixel to decide its belongingness to the focused area is an issue of relevance. This is due to the subjective, imprecise and not unique definition of quality. Therefore, it is unavoidable and considered as an issue of the fuzzy sets [39]. (3) This problem can be addressed by fuzzy image processing [16,40] because multi-focus images consist of imprecise detailed features due to the variations in illuminations, optical errors, focus and shooting angles.

As per the literature, every stage of image analysis consists of uncertainties and indeterminacies. Fuzzy sets can remove it with their powerful denoising ability. Thus, it is a continuous transformation of a crisp set to a fuzzy set [16-20]. Moreover, multi-focus images may suffer from uncertainty, noise and blurriness. Therefore, the fusion of the two partially focused parts is a challenging task. Thus, the fuzzy sets can deal with these indeterminacies and inconsistencies presented in the source images.

Furthermore, it removes the irrelevant noise and undesired background parts to maintain the relevant edges. Hence, it is considered as the most powerful approach to represent the vague, uncertain and imprecise features of multi-focus images. Thus, by this motivation, we have integrated the fuzzy method with CNN. Because CNN alone cannot handle the uncertainty and fuzziness present in the images, with the help of fuzzy sets, we can take those uncertainties present during the fusion of images. Therefore, the proposed method can address those challenges which are not handled by CNN alone.

Thus, no study is put forward by the amalgamation of the fuzzy sets with CNN to generate an FCNN method based on the analysis. Hence, a spatial domain MFIF method is proposed to integrate fuzzy sets and CNN to extricate the features of two partially focused source images. Fuzzy sets have been utilized in this research to quantify the indeterminacy and uncertainty of the image features in multi-focus images. CNN's are used to generate the fused images by extracting the high-level features from the fuzzified images.

Table 1 summarizes the performance of the state of art methods existing in the literature for MFIF. However, there were many shortcomings of the conventional MFIF based methods. There is a great need to evaluate based on metrics because quantitative score on various metrics assesses the fusion method's performance. The previous method's implementation was very time-consuming. Developing fusion rules and feature extractions require a lot of effort. Although images of good quality have been obtained, there is still a need to improve the performance of the MFIF methods. Moreover, the training of the dataset requires intensive time. Therefore, to overcome these challenges of implementing the MFIF methods, hybridization of the advantages of methodologies of fuzzy sets and convolutional neural network methodologies are done.

This study's significant contributions are as follows: i) A hybrid fuzzy convolutional neural network (FCNN) based MFIF method has been proposed where FS is integrated with CNN. ii) The FS has been used to automatically remove the various uncertainties presented in an image whereas, CNN has accomplished automatic features extraction. iii) In this study, the data-driven decision maps are obtained, overcoming the incapability of the manual generated decision maps by introducing the Siamese feature extraction module. It has been employed for the simultaneously salient feature extractions presented in the two given input images. iv) Generation of decision maps by using focus measure is the most challenging and crucial task that provides a focused region of the given input images and preserves those crucial regions in the fused image. Therefore, an extensive training dataset has been constructed based on the publicly available datasets. v) The proposed method has reduced the limitations faced by the traditional methods while designing the fusion rules and feature extractions. iv) The qualitative and quantitative analysis has been done on three datasets consisting of multi-focus images with both reference and non-reference evaluation metrics. This has been proved that the FCNN method attained superior results than the other six benchmark methods.

This study is organized as follows. In Section 2, the proposed methodology is elucidated. Result analysis is provided in Section 3. Finally, conclusions are concluded in Section 4.

2. Proposed methodology

The proposed method is formulated by combining fuzzy sets (FS) methodologies and convolutional neural networks (CNN).

2.1. Fuzzy sets

Zadeh introduced the notion of the FS in 1965 to handle imprecise information [41]. It is a mathematical expression that maps input data from a crisp set to a fuzzy set. It is a powerful tool to deal with vagueness, ambiguity, uncertainties, pixel distortions and blurredness present in the images due to poor illumination. The fuzziness and brightness distortion are unavoidable during the capture with sensors. Therefore, the fuzzy set approach has been used to determine and eliminate information loss. In addition, the FS provides effective outcome learning from uncertain data.

In comparison to others, the fuzzy approach is more accessible and less time-consuming. The usage of FS helps to remove the fuzziness and consequently improves the parameter results for quality assessment. Therefore, the problem of handling uncertainty has been overcome by incorporating the FS that help to deal with the presence of intermediate values with the assignment of a true membership degree ranging [0,1] to deal with ambiguous problems.

Consider a universe of discourse $Z' = \{z_1, z_2, \dots, z_N\}$. A mathematical representation of FS P is expressed in Eq. (1).

$$P = \{(z, \mu_P(z)) | z \in Z\} \quad (1)$$

where, $\mu_P(z)$ implies a membership function, whose value ranges between [0,1] such as $\mu_P(z): z \rightarrow [0,1]$.

Where z denotes the set universal element. The membership function is generally a degree of belongingness to FS. The value closer to 1 implies the complete belongingness of an element to FS, while 0 represents the non-belongingness to the FS. And then all the membership degrees of the element ' z ' are summed to get 1 as shown below.

$$\sum_{i=1}^p \mu_P(z) = 1 \quad (2)$$

Here, the number of fuzzy sets are given by p where z belongs.

The grayscale images consist of dark and bright pixels having a range between [0, 255]. So, the input images are mapped from the pixel domain to the fuzzy domain with the help of membership functions. Many membership functions such as triangular, bell-shaped, Gaussian and S-shape can generate image pixels. However, the selection of particular membership functions is user-defined [42].

In this research work, the S-type membership function is used for non-linear transform the pixel domain to the fuzzy domain. By this, unnecessary, uncertain information and vagueness of the image has been removed. Consequently, important information presented in the image has been highlighted, eventually producing an improved fuzzified image. Furthermore, a non-linear membership function known as S-type has been adopted because of the mentioned remarkable characteristics given below.

- A trapezoidal and triangular membership function has exhibited lower and upper levels belonging to grades 0 and 1, respectively.
- On the contrary, a non-linear membership function, i.e., S-shaped lower and upper level, represents the points 0.01 or 0.99, grades respectively.
- Another advantage is that the non-linear membership function, i.e., the S-shaped membership function, accurately represents the uncertainty of the fuzzy parameters. It modifies its shape with the parameter values.

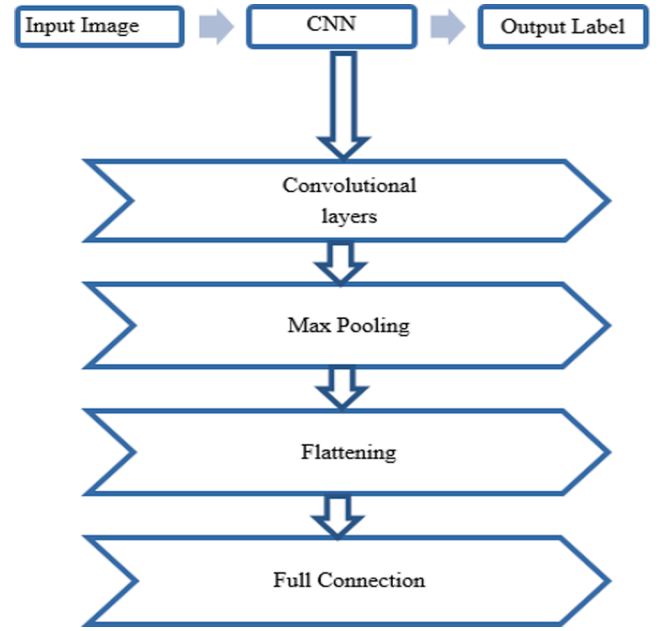


Fig. 1. Stages of CNN.

Based on repeated experiments, it has also been observed that metric values are better with the S-type membership function. Therefore, the S-shaped membership function is chosen for the fuzzification of the image pixels as described in Eq. (3).

$$\mu_{(P(i,j))} = \begin{cases} 0, & \text{if } 0 < \mu_{(P(i,j))} \leq \nu' \\ \left(\mu_{(P(i,j))} - \delta \right)^2 / (\chi' - \gamma')(\nu - \gamma'), & \text{if } \gamma' < \mu_{(P(i,j))} \leq \chi' \\ 1 - \left(\mu_{(P(i,j))} - \gamma' \right)^2 / (\nu - \chi')(\nu - \gamma'), & \text{if } \chi' < \mu_{(P(i,j))} \leq \nu \\ 1, & \text{if } \mu_{(P(i,j))} > \nu \end{cases} \quad (3)$$

where, $\mu_{(P(i,j))} \in [0,1]$ is the membership function of $P(i,j)$ that represents the number of noisy pixels. Now, images pixel values are mapped from the [0, 255] to [0, 1] representing the pixel fuzziness.

Let's take an image P for illustration. Its values range from H_{min} to H_{max} . $P(i,j)$ with pixel values are mapped to the domain of fuzzy characteristic. Inputs include three parameters (γ' , χ' and ν) and it's calculations are shown below.

1: Calculation of parameter γ' is done by calculating the entropy of an image as shown below.

$$\gamma' = Entropy(P) \quad (4)$$

2: The parameter χ' is obtained by computing the median value as shown below.

$$P : \chi' = median(P) \quad (5)$$

3: Compute parameter ν as:

- The maximum of the input source image is calculated by $M1 = \max(P)$.
- The mean of an input image is computed as $M2 = \text{mean}(P)$.
- Average is calculated as: $M = \frac{(M1+M2)}{2}$.

The degree of image intensity value ranges between 0 and 1, which was described by $\mu_{(P(i,j))}$. Therefore, the original pixel values [0, 255] are mapped to the fuzzy plane [0, 1]. Furthermore, if $\mu_{(P(i,j))}$ is 0, it indicates pixel $P(i,j)$ is noise-free. Thus, an original pixel is given as output without any need for filtering having a detailed image. If $\mu_{(P(i,j))}$ is 1, it implies pixel $P(i,j)$ pixel consists of noise which further requires image

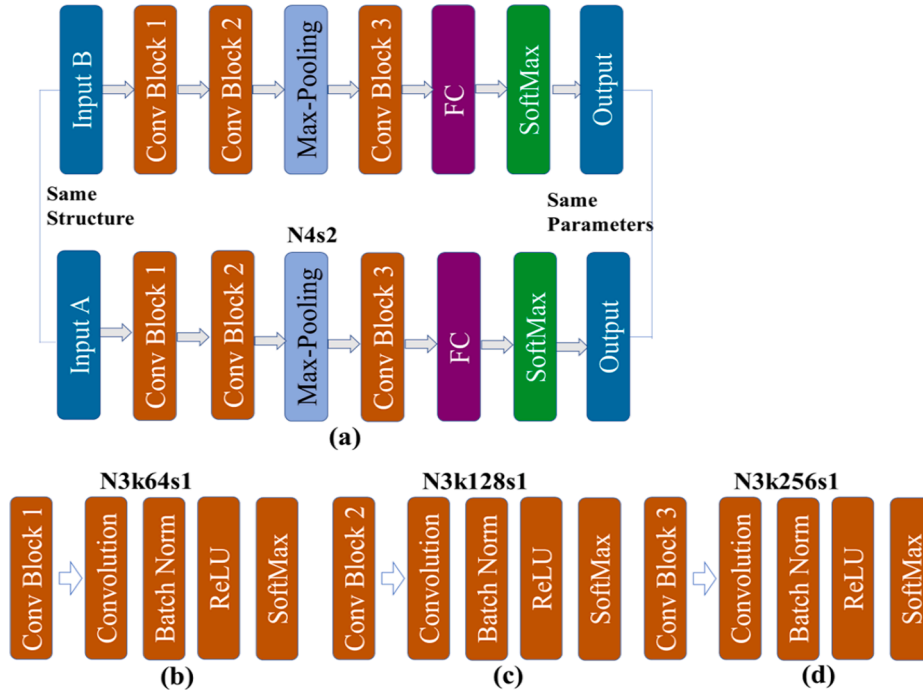


Fig. 2. Detailed Siamese CNN architecture. (a) Complete network. (b) Description of the Conv Block 1. (c) Detailed Conv Block 2. (d) Provides description of Conv Block 3.

filtering. If membership grade is between $0 \leq P(i,j) \leq 1$, it represents that the current pixel is contaminated with the noise which will be eliminated using Eq. (3). χ' is another index that prevents erroneous prediction when $P(i,j)$ is unable to identify the noise. The values of indicators such as $\mu_{(P(i,j))}$ and M are high in the presence of noise and vice versa. So, this process has generated improved enhanced images, which are very useful for image fusion. Now, these images are employed to represent the more detailed information.

2.2. Siamese convolutional neural network architecture

ConvNet or CNN is a supervised deep learning model designed to learn a hierarchical representation of features for the data presented in an image with various levels of abstraction. Fig. 1 describes the CNN framework having an input image with the convolution layer, max-pooling layers, flattening of the features, and an output indicating a label vector with the help of full connection. In addition, it defines the probability depending on the categories of features. Each stage has a feature map (FM) parallel to abstractions for the particular image pixel features [43].

The primary purpose of CNN is to find features in the image by feature detector and put it into a feature map. There is an increasing development in visual quality, which describes the promising prospective for extracting features in CNN [44]. The max-pooling is used as a data reduction. The benefits of max-pooling are that features remain preserved. In the convolutional operation, X_i, H_j indicates the i_{th} input FM and j_{th} output FM. The rectified linear unit (ReLU) which is a non-linear activation function and convolution determined by the equation given below.

$$H^j = \max \left(0, b^j + \sum_j P^{ij} * X_i \right) \quad (6)$$

where P^{ij} indicates the convolutional filter of X_i and H^j respectively and bias is indicated as b^j . Convolutional operation is indicated by $*$ symbol. Commonly, max-pooling is utilized for this work which is given by:

Table 2

The Siamese CNN architecture of the proposed FCNN method of one path.

#	Type of Layer	Filters	Size	Input	Output
1	Conv Block 1	64	$3 \times 3/1$	$64 \times 64 \times 3$	$64 \times 64 \times 64$
2	Conv Block 2	128	$3 \times 3/1$	$64 \times 64 \times 64$	$32 \times 32 \times 128$
3	Max-Pooling	128	$2 \times 2/2$	$32 \times 32 \times 128$	$16 \times 16 \times 128$
4	Conv Block 3	256	$3 \times 3/1$	$16 \times 16 \times 128$	$8 \times 8 \times 256$
5	FC	512	$3 \times 3/1$	$8 \times 8 \times 256$	$8 \times 8 \times 512$

$$Y_{r,c}^i = \max_{0 \leq m,n < t} \left\{ X_{r,c+t,n}^i \right\} \quad (7)$$

$Y_{r,c}^i$ indicates the location of neurons at (r, c) in the output FM. The i_{th} input map has a region of size $t \times t$ in FM of X_i . It also prevents the loss of information. Here, m indicates input feature map and n indicates output feature map that constitutes of kernel with size $D \times D \times m$. Where $D \times D$ is the local receptive fields. Each constituent of the kernel present in the image owns a bias. Then the fully connected layer is used to concatenate the different features extracted in the previous layers. Next, the flattening of the pooling layers was done to get so many pooled feature maps. Finally, the output neurons present in the last layer assigns a probability whose value limits between 0 and 1.

Fig. 2 illustrates the Siamese CNN architecture of the proposed method. The Siamese network is utilized to share the same weights between two different branches. One branch deal with image A while the other handles image B. Subsequently, to extract the crucial features from the input images, in Conv Block 1, a filter of size 3×3 has been applied followed by batch normalization (BN') and ReLU function. Here, N3k64s1 implies the filter (N) of size 3×3 , number of feature map (k) 64, with the stride (s) of 1. Likewise, N3k128s1 and N3k256s1. Lastly, fully connected (FC) and SoftMax operations are utilized to generate the weighted maps whose range is $[0, 1]$. In summary, the first three Conv Blocks have generated 64 FM after applying 64 filters. In contrast, 128 FM are extracted by Conv Block 2 with the use of 128 filters. Further, the extracted 256 FM is fed to the fully connection to integrate it with 256-dimensional vectors. Thus, a 2D output vector is produced. Finally, the

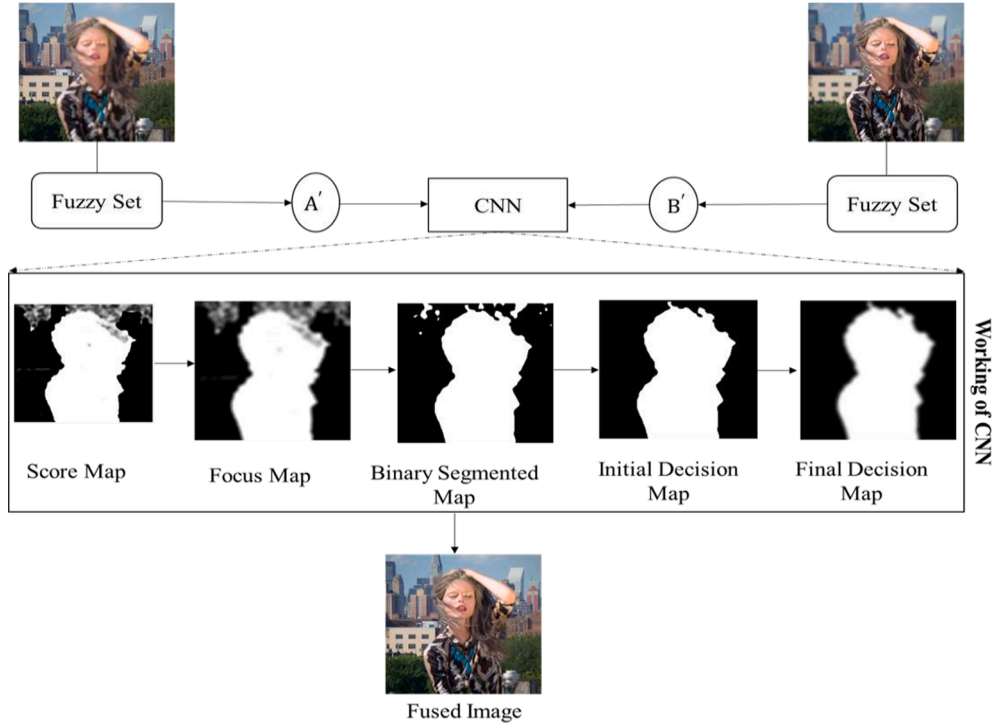


Fig. 3. Schematic block diagram of the FCNN method.

SoftMax function calculates the probability distribution among two classes. Hence decision maps are generated by the automatically extracting the feature from the two given images.

Table 2 lists the tabular representations of the layers used for the proposed FCNN method. The three layers of CNN have been used. The first Conv Block has produced 64 feature maps by applying 64 filters. The second Conv Block has given 128 FM after applying 128 filters. CNN has automatically applied these filters due to their self-learning property for feature extraction. The time computational complexity of the FCNN is also acceptable. It is used in two parallel paths, and one is for an image in which the front-end of the image is clear and the second one in which the back-end part of the image is clear. Three convolutional layers have been used for detecting features. The size of the feature detector was 3×3 and chose stride 1. Afterwards, FC is input with 256 FM. And then 2-D output is attained by combining with 256-2D vectors. The two paths have been used during CNN work. A SoftMax layer has produced the probability distribution over two classes.

2.3. Proposed MFIF implementation with fusion framework

This section gives an in-depth explanation of the proposed FCNN method for the MFIF. An attempt is made to handle the problems of CNN by introducing an integrated FCNN method to obtain an image having all its characteristics. The schematic block diagram of the proposed FCNN based MFIF framework is depicted in Fig. 3. The FCNN method constitutes the following steps. First, a modified fuzzy set is applied to convert the grayscale images into the fuzzy domain to remove the uncertainties like noise, information loss and distortion of an image. Then, Siamese CNN is used to identify the focused and unfocused parts as a decision maps.

2.4. MFIF framework

Two grayscale images A and B are given as input to the pre-trained CNN model. Initially, the two-source images are fuzzified as A' and B'. Then, these images have been passed to the designed Siamese CNN framework to recognize the image pixels presented in the multi-focus

images to compute the focused and unfocused pixels as a decision map. The Siamese CNN has performed this process of segmentation of the focused and unfocused regions and then these focused and unfocused regions have been reconstructed by the developed reconstruction approach.

The process of fusion is done by giving 2 fuzzified images to the trained network. The network training has been done with a patch size of 16×16 generates a score map SM. SM represents the extracted focused information from a set of patches of (16×16) size. Its value ranges from [0, 1]. A value closer to 1 (white) and 0 (black) implies obtaining more detailed focused information. The obtained size of SM is expressed below.

$$SM = \left(\left\lceil \frac{ht}{2} \right\rceil - conv_patchsize + 1 \right) \times \left(\left\lceil \frac{wt}{2} \right\rceil - conv_patchsize + 1 \right) \quad (8)$$

If $0 < SM < 1$, it implies the focused pixel regions. Height and width are denoted by ht and wt , respectively. It is observed from the experimental analysis that the size of the SM is reduced to half of the original image size. The reason is the overlapping of the pixels. Consequently, ht , wt and $conv_patchsize$ are reduced to half resulting in 8×8 $conv_patchsize$. Therefore, the overlapping pixels of the SM are mapped back to the original size by the averaging method. Subsequently, a FM is generated precisely the size of the source images. The correct focused regions are obtained where the pixels 0 (black) or 1 (white) give more excellent image detail. Moreover, redundant pixels represent the plain regions (gray) having a pixel value of 0.4.

$$P(FM) = Black \cdot \delta(FM - 0) + Gray \cdot \delta(FM - 0.4) + White \cdot \delta(FM - 1) \quad (9)$$

where, $\delta(x - x_0)$ is Dirac-Delta function.

The segmentation [45] of FM is done by taking the threshold of 0.4 to have a binary segmented map as described in Eq. (9). A threshold of value 0.4 has been chosen empirically to generate an accurate focused map known as the focused map quality control factor. Several images have been tested to observe the quality of generated binary maps and fused images. The iteration of a reasonable value has carefully chosen the threshold factor from 0 to 10. It is observed that the clarity of the

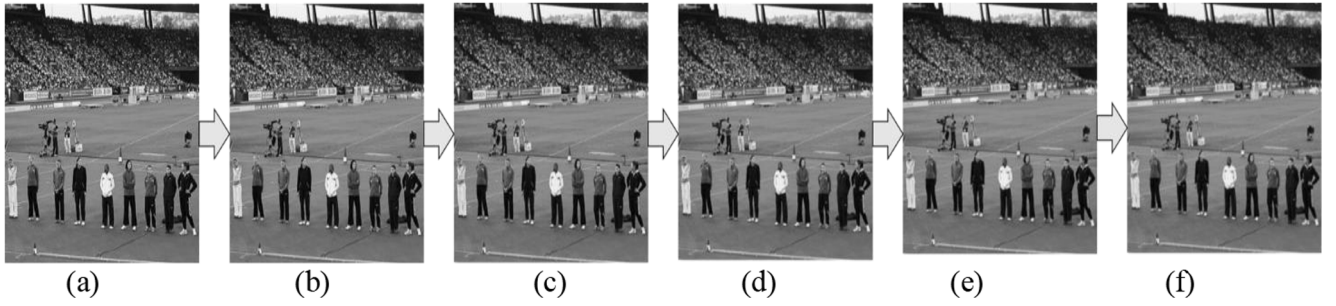


Fig. 4. Blurred images of different levels reproduced by Gaussian filter (a) Original image (b) Blurred Image (level 1) (c) Blurred Image (level 2) (d) Blurred Image (level 3) (e) Blurred Image (level 4) (f) Blurred Image (level 5).

generated fused image has been decreased with a threshold factor of less than 0.4. On the other hand, if the value is increased to more than 0.4, the algorithm takes more computational time to produce a fused image with less detailed information. Therefore, to balance good quality and computationally expensiveness, a threshold factor of 0.4 has been selected by experimenting with the optimal values. This threshold factor produces the best binary segmentation maps while giving higher performance metrics results than the other threshold factors.

The proposed FCNN MFIF method is described in Algorithm 1.

Algorithm 1 Proposed FCNN algorithm.

Input: Image A and Image B .

Step 1: Read two images A and B .

Step 2: Fuzzification

Step 2.1 Get the fuzzified images A' and B' by transforming the input source images A and B from spatial domain to fuzzy domain with the FS.

Step 2.2 Input these resultant fuzzified images to the Siamese CNN.

Step 3: Focused and unfocused pixel regions detection by Siamese CNN.

Step 3.1 For every quantified pixel in images, A and B are fed to pre-trained CNN.

Step 3.2 Compute SM after the input to the CNN. The size of SM is given in Eq. (8).

Step 3.3 After determining the SM , generate the FM using the average method given as in Eq. (9).

Step 4: Binary segmentation.

Step 4.1 Generate the binary segmented map after calculating FM by Eq. (10).

Step 4.2 Use the weighted average (WA) method to determine the ID' map.

Step 5: Final decision map.

Step 5.1 Determine the final decision map after obtaining the initial decision map by using GF by setting the following parameters: $r = 7$ and $e = 0.2$.

Step 6: Multi-focus image fusion.

Step 6.1 MFIF is performed by using the pixel-wise rule.

Step 6.2 For the generation of the final fused all in-focus image, Eq. (13) is calculated.

Output: Fused image F .

$$P(x, y) = \begin{cases} 1, & FM(x, y) > 0.4 \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

Fig. 3 highlights the accurately classified gray pixels of the binary map. It is visible on a white background. Now, it constitutes the image part which is near to 0 or 1 values. However, the acquired binary map is contaminated with unwanted small objects or noise. Therefore, a small region removal approach is utilized to eliminate the misclassified elements of the FM . Hence, a precise noise-free initial decision map (ID') is generated with the use of 'bwareaopen' as represented in Eq. (11).

$$ID' = \text{bwareaopen}(BM', \text{area}) \quad (11)$$

Here, the threshold value of the area was adjusted to 0.01 as shown in Eq. (12):

$$\text{area} = 0.01 \times ht \times wt \quad (12)$$

Here, ht and wt indicate the length and breadth of the input source images. The undesirable artifacts present on the edges are removed from the initial decision map using the guided filter. After getting the final decision map, the final resultant output is obtained with the pixel-wise weighted average technique, as given in Eq. (13).

$$F(x, y) = FD(x, y)A'(x, y) + 1 - FD(x, y)B'(x, y), \quad (13)$$

where A' and B' are the input source images. Decision map is denoted by $FD(x, y)$ and $F(x, y)$ is the fused output image.

3. Results analysis

In this section, both qualitative and quantitative evaluations have been carried out among proposed FCNN and other MFIF techniques, namely NSWT [11], GF [15], CNN [22], ECNN [33], IFCNN [36] and DRPL [37]. Various statistical parameters have been computed, viz. mutual information (MI), edge information (EI), structural similarity (SS), peak signal-to-noise ratio (PSNR), root mean square error (RMSE) and human perception (HP) [1]. The greater metrics values except RMSE indicated the higher quality of fused image generation, whereas RMSE having a low value implies less distortion obtained in images. A thorough discussion, as well as examination of FCNN outcome, is discussed below.

3.1. Datasets acquisition

This research work includes three datasets, namely dataset 1, dataset 2 and dataset 3. Dataset 1 contains pair of 20 'lytro' multi-focus images that are publicly available [1]. Dataset 2 contains 21 sets of multi-focus images. Dataset 3 consists of 15 sets of input and reference multi-focus images [46,47]. The size of each image in a dataset is 520×520 pixels. The FCNN method has been implemented in MatlabR2019a 64-bit, Intel® Core i7-9700F 3.00 GHz and graphic card is NVIDIA GTX 1660 and 16 GB RAM.

3.2. Training dataset

This is the foremost crucial and complex task to train the supervised CNN due to the requirement of the labeled training image sets. Moreover, no datasets are publicly available and MFIF with CNN demands many ground truths labeled images. Adopting a manual way to mark the vast number of images is a computationally intensive and impossible task to achieve. According to the literature study, some authors like Tang et al. [48] developed CNN-based MFIF using CIFAR-10 database images. In this, dataset images were considered to be focused images having a resolution of 32×32 . A dataset construction was done by labeling the different blurred levels and adding the unfocused masks. This procedure of the training dataset has some disadvantages: clarity of these dataset images was not good compared to multi-focus images and the masks were constructed manually with more complexity. On the other hand, Liu et al. [22] constructed a training dataset by inputting the focused images (from the ILSVRC 2012) and created unfocused images by blurring the images simulated by the Gaussian filter. Although this dataset construction was simple but attained focused and unfocused regions did not appropriately recognized.

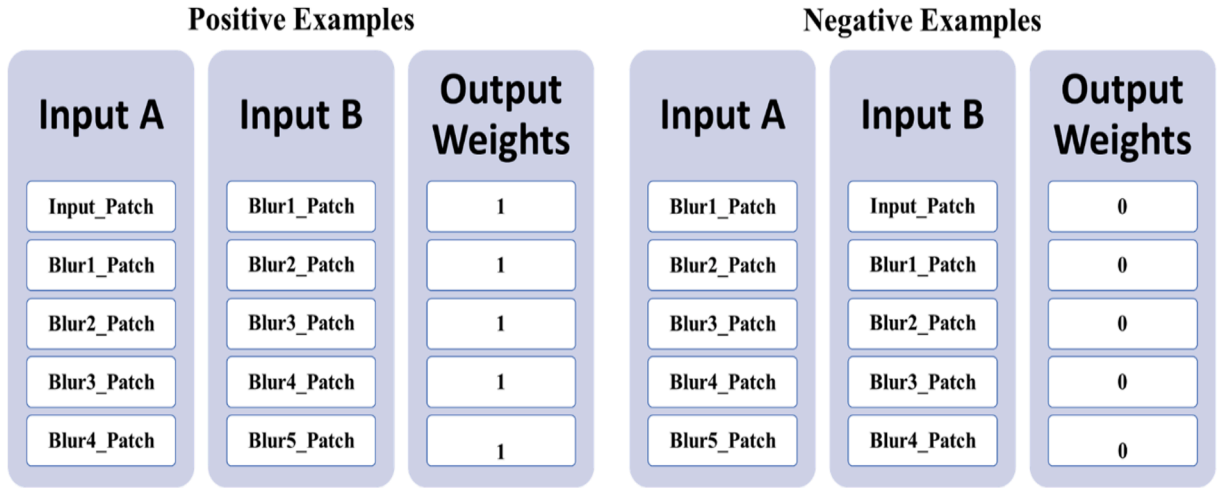


Fig. 5. Selection criteria for MFIF dataset.

Table 3

Comparison of the selection of patch size.

Patch size	Max-pooling layers	Accuracy
8×8	Less	Poor
16×16	Less	Good
32×32	Too large	Good accuracy but network become too complex due to increase in the number of max-pooling layers

Therefore, in this research, training examples are generated using the publicly accessible ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2012 dataset and these images are considered to be focused. Due to the unavailability of the referenced images, natural multi-focus blurred images are created by adding Gaussian blur after processing with Gaussian filters with a standard deviation of 0.5. Fig. 4 presents the original images and the blurry images with five different levels. Blurred images are constructed with five different levels. In the first level, the simulation uses a Gaussian filter with a 0.5 standard deviation value. And in the second level, the blurred image has been obtained using the first level blurred image with the Gaussian filter and the same procedure has been continued till five levels. Finally, selection criteria illustrated in Fig. 4 are employed for the given input patches extraction from the focused and unfocused images.

Fig. 5 illustrates the patch-based selection strategy. It detects focused and unfocused pixels to produce the image patch, subsequently giving input to the training dataset. This approach reduces the time that could be impossible with detecting focused and unfocused pixels from the neighboring information of the pixels, which is a time-intensive task. So, the patch size was chosen to be 16×16 , as explained in Table 3, it has given the comparison for the choice of patch size. In the proposed work, 16×16 patch block has been chosen. It can be justified as when patch size is set to be 8×8 , classification accuracy achieved is not so good because it is too small in size to train the CNN model. When the patch block size is 32×32 chosen, good accuracy has been achieved, but the drawback is that the number of max-pooling layers has increased in this case. Hence, it became difficult to calculate the number of max-pooling layers, thereby making a complex network. Therefore, it is good to select a 16×16 block of an image to train the CNN model.

Henceforth, a total of 5,000 images from an ImageNet database are retrieved [49], where 2500 images are positive samples, whose output is set to be 1 and 2500 negative samples with 0. Each sample consisted of two input image patches (16×16) and reference image labels, i.e., 1 or 0. Image patch is given as an input to the training dataset. Each image is cropped into patches of size 16×16 without overlapping using bicubic

transformation. CNN model has been trained with CAFFE framework with 16×16 patch images [50]. Testing has been done by MatConvNet [49] and weights have been initialized by using the Xavier algorithm [51].

The updating of weights has been done by Eq. (14).

$$\alpha_{i+1} = M \cdot \alpha_i * W^* \gamma \cdot w_i^* \gamma^* \frac{\partial L}{\partial W_i} \quad (14)$$

Here, momentum variable is defined by α , iteration index is represented by i , 'L' implies the loss function and M denotes momentum fixed at 0.9. Weight decay is given by W set at -0.0005 and γ which is set at 0.002 defines to be a learning rate. The derivative of L is given as $\frac{\partial L}{\partial W_i}$, where w_i denotes a weight. The γ with 0.002 is chosen. The learning rate with a higher value adversely affects the calculated loss, while the lesser learning rate suffers from an increased epoch time. Therefore, weight decay is kept at 0.0005 after the various experimentations. The colored images are converted to a grayscale domain. Images are rotated by 90° and 180° in both horizontal and vertical directions. Data augmentation is done to increase the dataset size.

3.3. Performance evaluation metrics

Objective assessment is vital in image fusion as the quantitative score on various metrics assesses the fusion method's performance. In addition, the evaluation metrics are used to investigate the implementation of the FCNN method. Both reference and non-reference metrics are employed.

3.3.1. Non-reference metrics

The term non-reference indicates the non-requirement of any reference image to evaluate the performance of the fused images. The various utilized non-reference metrics are as follows.

3.3.1.1. MI. MI measures the transfer of salient information from the two input images to the resultant single fused image. The proposed method performance has been tested using this metric. It can measure the features and visual information passed from two input images to fused images. MI correlates better with the subjective quality of an image. The fused image with a greater MI value indicates a better-quality image.

$$MI = \sum_{f,a} \mathbb{E}_{A'F'}(f,a) \log \frac{\mathbb{E}_{A'F'}(a,f)}{\mathbb{E}_{A'}(a)\mathbb{E}_{F'}(f)} + \sum_{f,b} \mathbb{E}_{B'F'}(f,b) \log \frac{\mathbb{E}_{B'F'}(b,f)}{\mathbb{E}_{B'}(b)\mathbb{E}_{F'}(f)} \quad (15)$$

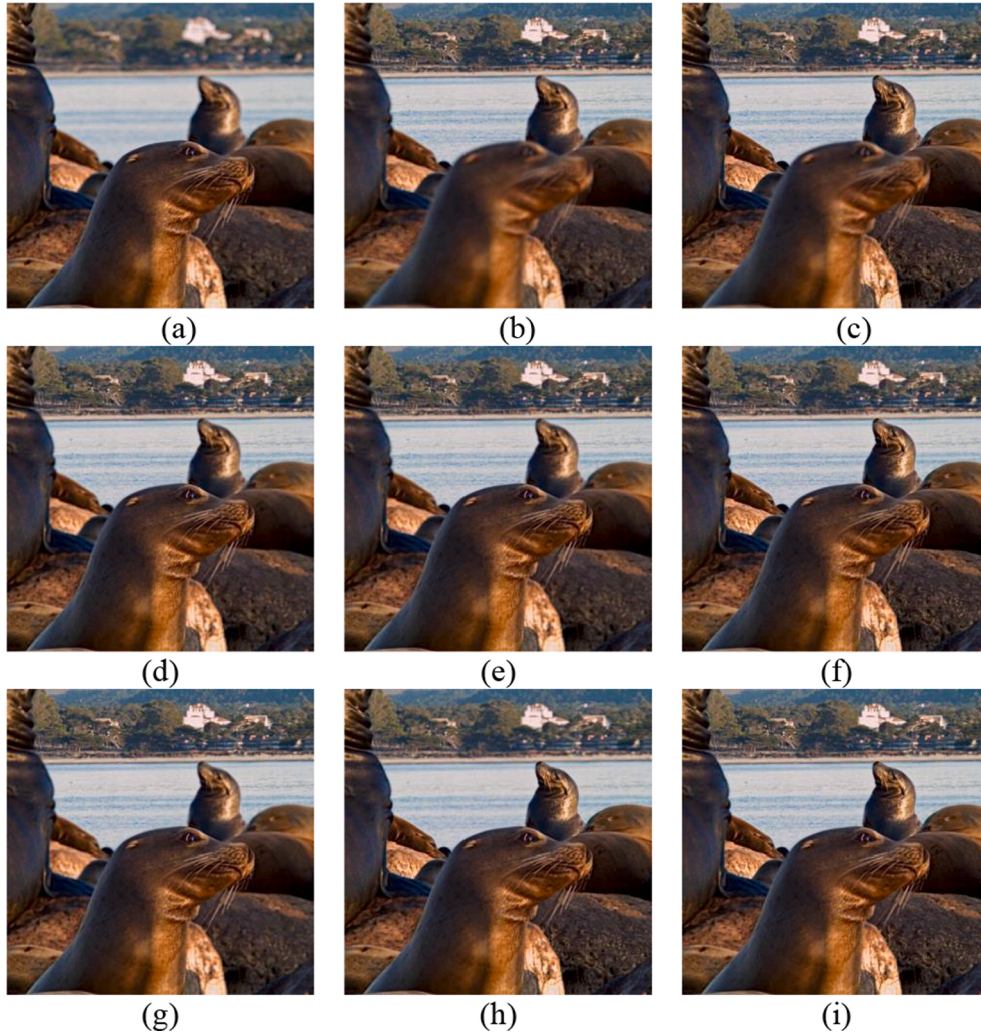


Fig. 6. Multi-focus image set. (a) Source image1. (b) Source image 2. (c) NSWT fused output. (d) GF fused output. (e) CNN fused output. (f) DRPL fused output. (g) ECNN fused output. (h) IFCNN fused output. (i) FCNN fused output.

where A' and B' indicate input images, fused image is given as F' . $E_{A'F'}$ and $E_{B'F'}$ defines the joint histograms of all images. While $E_{A'}$, $E_{B'}$ and $E_{F'}$ represent the corresponding histograms of all images i.e., A' , B' and F' .

3.3.1.2. EI. It transfers significant information (visual/edge) transferred from the source input image to the final image known as fused image. It is evaluated by Eq. (16).

$$EI = \frac{\sum_{s=1}^S \sum_{t=1}^T (Q^{A'F'}(i,j)W^{A'}(i,j) + Q^{B'F'}(i,j)W^{B'}(i,j))}{\sum_{i=1}^S \sum_{j=1}^T (W^{A'}(i,j) + W^{B'}(i,j))} \quad (16)$$

Here, input and fused images are represented as A' , B' and F' , respectively. $W^{A'}(i,j)$ and $W^{B'}(i,j)$ are the pixels weight. $Q^{A'F'}(i,j)$ and $Q^{B'F'}(i,j)$ represents the similarity of two images A' and B' . (i,j) are the pixel locations.

3.3.1.3. SS. It is used to measure the preservation of structural information into a final fused output image. The output value approaching 1 or 0 indicates the high/low-quality image.

$$SS = \sum c(w')(\alpha(w')Q_0(A', F'/w') + (1 - \alpha(w'))Q_0(B', F'/w')) \quad (17)$$

Here, A' and B' are the two given images, fused image is represented by F' . $\alpha(w')$ denotes a local weight with $0 \leq \alpha(w') \leq 1$. A value for SS close to 1 implies the less structurally distorted image with more

important features transfer in the fused image. $c(w')$ computes weights. w' is the window size.

3.3.1.4. HP. This metric is based on the perception of humans on the quality of IF. Contrast sensitivity filtering (CSF) has been used to filter the input and fused images. Then, the contrast preservation map (CPM) was computed. Finally, the quality map has been obtained.

$$HP = \alpha_{A'}(i,j)Q_{A'F'}(i,j) + \alpha_{B'}(i,j)Q_{B'F'}(i,j) \quad (18)$$

where $\alpha_{A'}(i,j)$ and $\alpha_{B'}(i,j)$ denote the saliency maps. $Q_{A'F'}(i,j)$ and $Q_{B'F'}(i,j)$ determines the similarity among A' , B' and F' .

3.3.2. Reference metrics

The proposed method evaluation is performed by comparing the fused image with the reference image. This is a very effective approach to analyze the efficiency of the fusion methods. Therefore, 2 well-known metrics such as peak signal-to-noise ratio (PSNR) and root mean square error (RMSE) are employed. The used metrics are described below.

3.3.2.1. PSNR. It evaluates the similarity between the reference and the final fused image. Higher values indicate the better fusion results. Let R'_{ij} indicates reference image and final fused image is represented by F'_{ij} . It is calculated by Eq. (19).

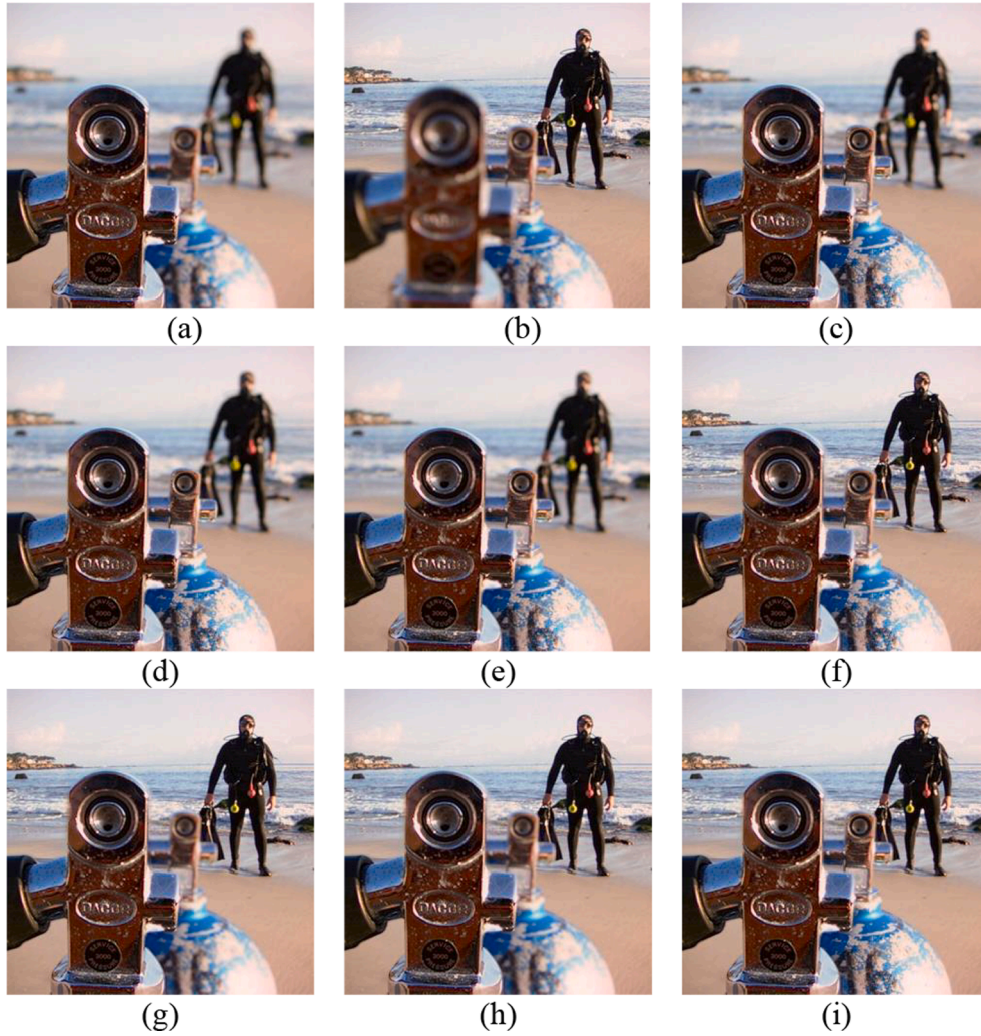


Fig. 7. (a) Far focused input image 1. (b) Near focused input image 2. (c) NSWT fused output. (d) CNN fused output. (e) GF fused output. (f) DRPL fused output. (g) ECNN fused output. (h) IFCNN fused output. (i) Proposed FCNN output.

$$PSNR = 10 \log_{10} \frac{\sigma^2}{\frac{1}{A' \times B'} \sum_{i=1}^{A'} \sum_{j=1}^{B'} (\mathfrak{R}_{ij}' - F_{ij}')^2} \quad (19)$$

3.3.2.2. RMSE. It calculates the changes per pixel because of the image fusion methods [52]. It is represented using Eq. (20).

$$RMSE = \sqrt{\frac{1}{A' \times B'} \sum_{i=1}^{A'} \sum_{j=1}^{B'} (R_{ij}' - F_{ij}')^2} \quad (20)$$

3.4. Result analysis of proposed FCNN method

Numerous pairs of input images are used for the examination of the feasibility of the FCNN method. Fig. 6 and Fig. 7 show the difference in the fusion results compared to the other competing methods. Firstly, the comparison has been made based on visibility. Fig. 6 depicts the set of multi-focus images. These images have been used to compare NSWT, CNN, GF, DRPL, ECNN, IFCNN and the proposed FCNN method. The input images are depicted in Fig. 6(a) and (b). These are the near focused and far focused images. The fusion output of the image for the NSWT, CNN, GF, DRPL, ECNN, IFCNN and FCNN methods are illustrated in Fig. 6(c) to Fig. 6(i), respectively.

The fused image obtained in Fig. 6(c) and Fig. 6(h) by the NSWT and IFCNN methods produced blurredness with poor visibility. Moreover, edges are not sharp, having uneven boundaries giving unclear images.

Thus, inferior results indicated that these methods have poor directionality. On the contrary, the fused images presented in Fig. 6(f) and Fig. 6(g) is superior to the NSWT and IFCNN methods. A better-quality image is obtained, but there is still edge interruption and over smoothing issues. The image given by the CNN method is of low contrast. It can be observed from Fig. 6(e) only the front part of the input image is visible with little ambiguity. At the same time, the rest of the region consists of blurriness due to the overshadowing. The fused image generated by the GF is better than the CNN method. It produced better edges and boundaries than CNN. But all the objects are not clear. The image obtained using the FCNN method has given fine details with relatively accurate edges differentiating focused and unfocused parts. Moreover, the front and back parts of the image, as shown in Fig. 6(i), are clearly compared to the other existing MFIF methods. It can be analyzed that the proposed method is robust towards the noise. It has removed the blurred information and generated the final fused image. Thus, the FCNN method has given the improved fused image.

Another example has been shown as a fused resultant image using different MFIF methods. Fig. 7(a) and Fig. 7(b) depict the multi-focus source input images. The fused output image given by different MFIF methods has been illustrated in Fig. 7(c) to 7(i). From the Fig. 7(c), 7(d) and 7(e), it has been observed that it has generated the blurred image. It has not fused any focused regions from Fig. 7(b) into the final fused image. Hence, it resulted in undesired fused images. Nevertheless, ECNN and IFCNN attained better fused image than the NSWT, GF and CNN.

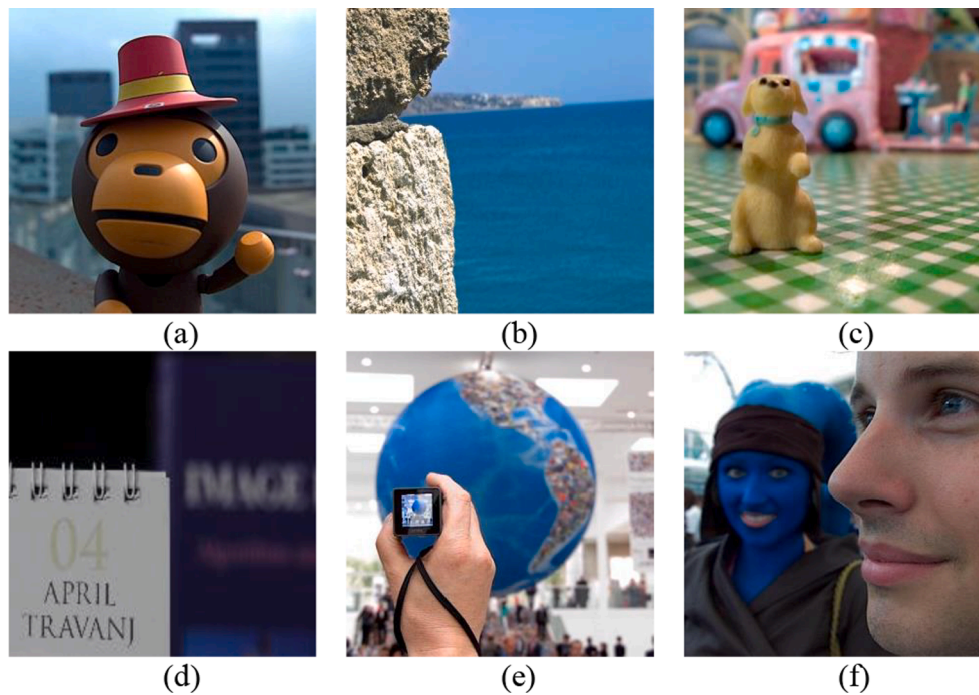


Fig. 8. Six near focused input images.

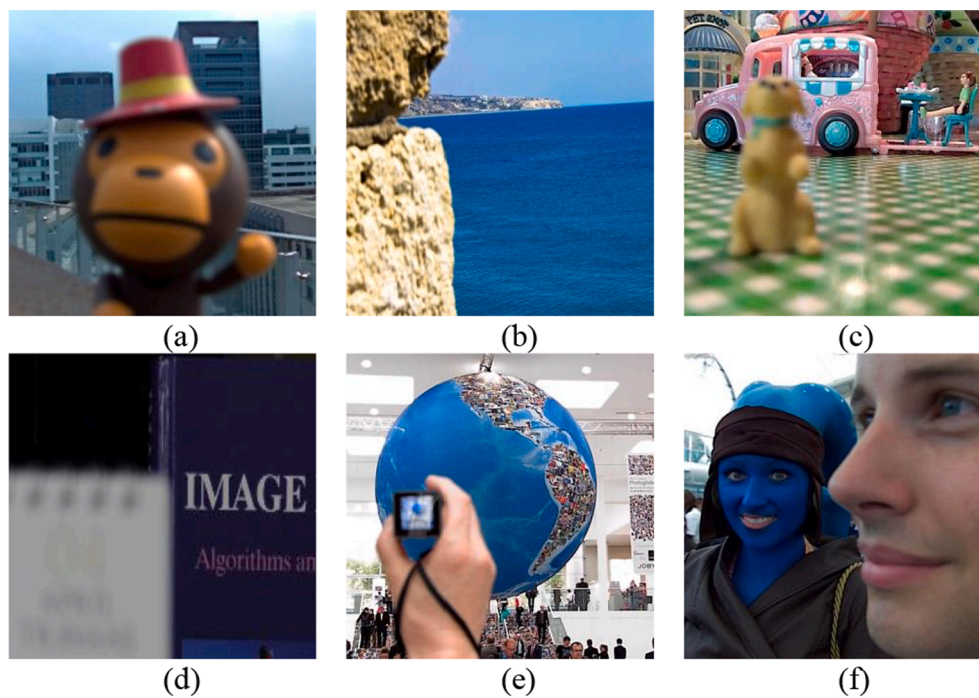


Fig. 9. Six far focused input images.

However, it is limited to fuse only some regions correctly. It cannot combine the objects from Fig. 7(b), hence giving a blurry appearance.

The DRPL method is better than the above-discussed methods but cannot correctly merge all the focused regions into the final fused image, as shown in Fig. 7(f). Thus, it suffers from clarity and contrast issues. The proposed FCNN method successfully combined the partially focused images. It is best in appearance in comparison to the other competing methods. Accurate, sharp edges with good visibility having the appropriate boundary is provided by the fused FCNN method. Furthermore, a good contrast image with maximum brightness is generated by the

FCNN method. Therefore, the proposed FCNN method has systematized the focused and unfocused pixels. Thus, it can be concluded that the FCNN method has better resultant fused images than the other methods.

The FCNN method efficiency is further validated by using different multi-focus images as presented in Figs. 8 and 9 depicting the multi-focus source input images. The fused output image given by proposed methods has been illustrated in Fig. 10. It can be seen that the image produced by the FCNN method is clear in vision.

Fig. 11 presents the quantitative evaluation of the proposed FCNN method on 41 multi-focus images. The individual results for images

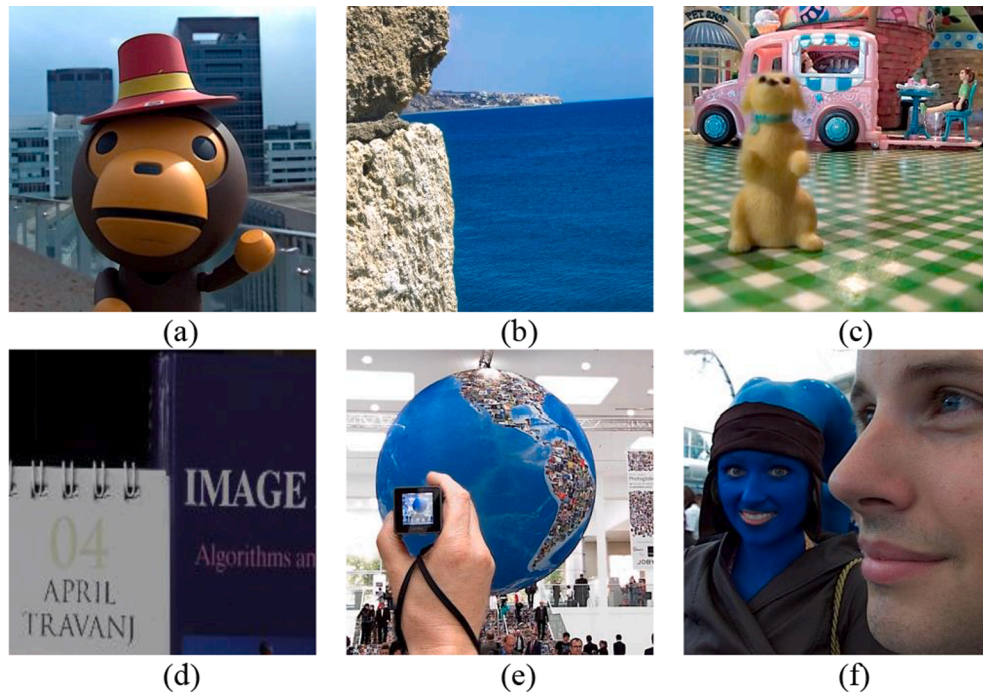


Fig. 10. Six fused output images by FCNN.

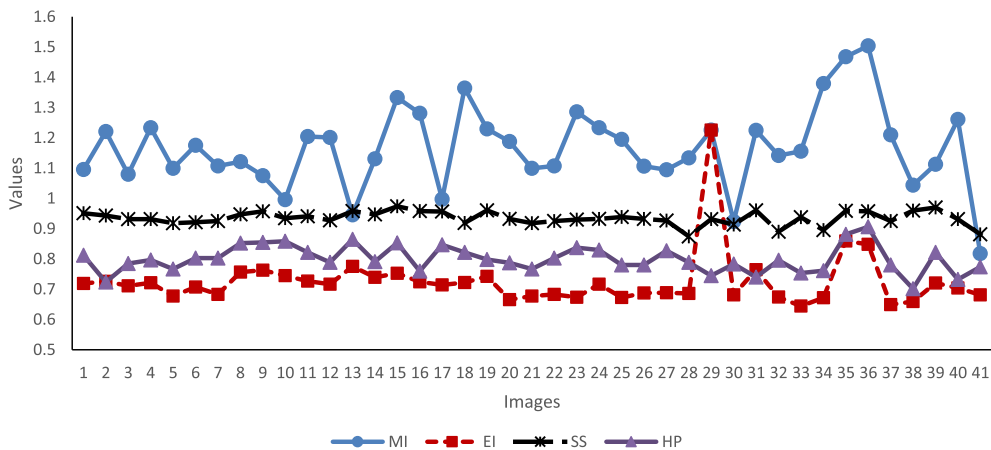


Fig. 11. Quantitative evaluation of proposed FCNN method.

using various evaluation metrics are listed. The average value of MI is 1.1676, EI is 0.7281, SS is 0.9350 and HP is 0.8020. It has been noted

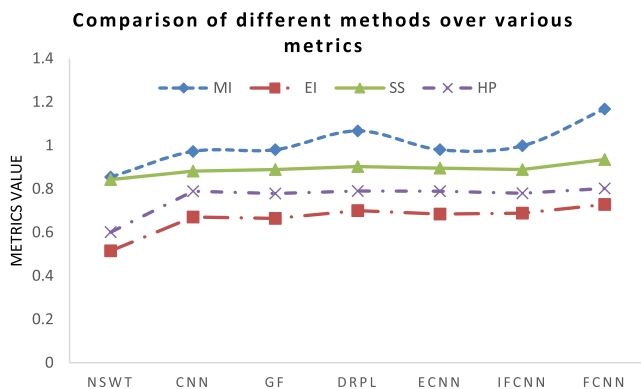


Fig. 12. Average comparison of the different methods.

that 'Img 18' has the highest MI value (1.3642) and the 'Img 41' has the lowest MI value (0.8180). It has been noted that the image with the highest MI value has more clarity and visibility. It contains more contents of both source input images. Moreover, the boundaries of the image are clear. It has been evidenced that the image with the lowest MI value has less clarity. The visibility of the image is not so good, having weak edges. The greater metric values imply the better quality of fused output image. In 'Img33', it has been found that the lowest EI value (0.6440) stated less information of the source image and image boundaries with little blurredness in less accurate edges. The highest EI value (0.8593) has been attained by the 'Img 35', which has the best visual effects. It has a sharply focused image with clear boundaries. Even the small parts of the image have an excellent visual appearance.

The SS metric measures the structural information presented in the resultant fused image from the given input images. Its range is defined between [0,1]. The greater value implies the better resultant fused image. Therefore, a value for SS close to 1 implies a more structured preserved image is generated. It is observed that the 'Img 15' has achieved the best value (0.9737), which describes both parts of the image

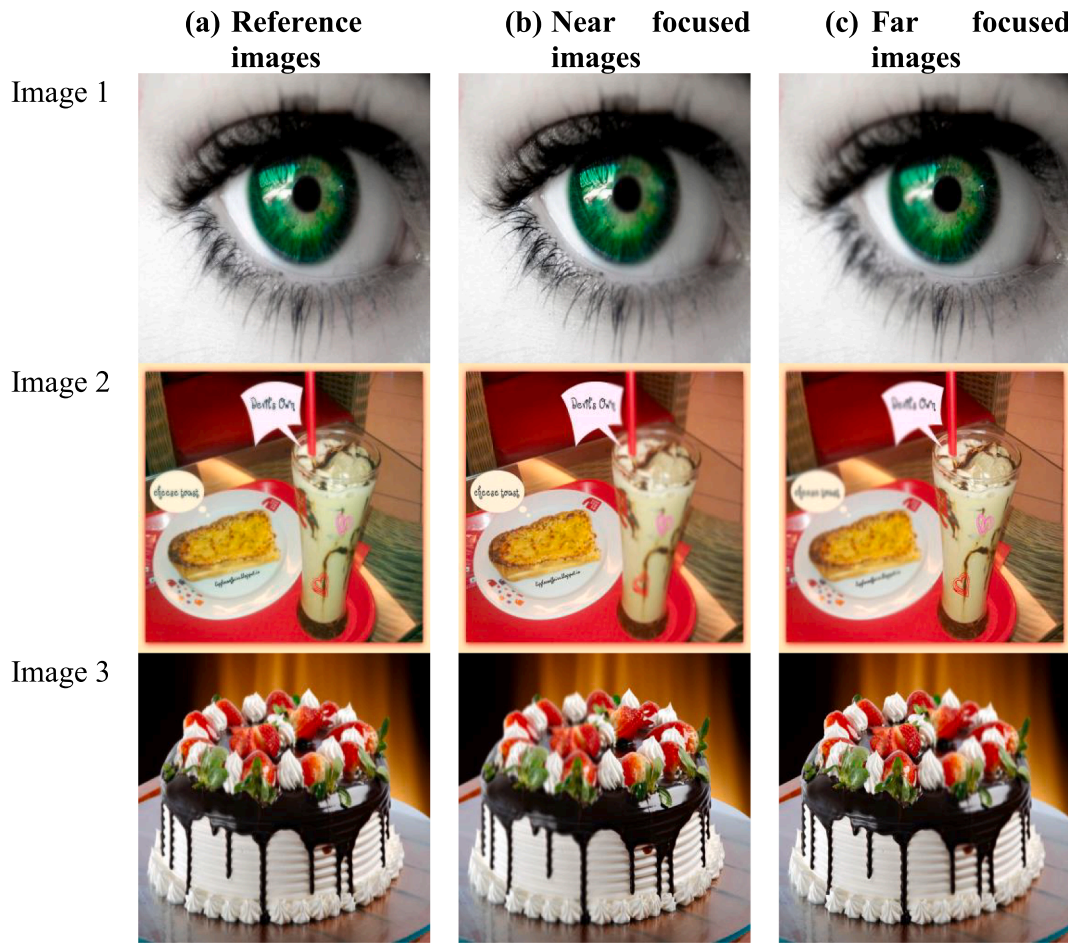


Fig. 13. (a) Reference images. (b) Left focused input images. (c) Right focused input images.

effectively. The accurate boundaries and good contrast image have shown the best visibility of the image. On the contrary, although the 'Img 28' has contained the characteristics of both source input images, the small necessary information of the image is not visible. It has been observed that the 'Img 35' has achieved the highest HP value (0.8811), giving the focused image whereas, the 'Img 38' has the least HP value (0.7015). The image has shown a good appearance but having an edge breakage issue as well as unclearly differentiating the focused and unfocused part in the region.

The comparison of average values of each quantitative assessment of the proposed and the other existing methods based on each metric (MI, EI, SS and HP) is illustrated in Fig. 12. In Fig. 12, a comparison of average values of 41 different multi-focus images with other methods has been done for the validation of the effectiveness of the proposed FCNN method. It can be analyzed that the FCNN method has acquired the most extensive MI (1.467) among the other competing fusion methods. It indicates that the most considerable information of the two-source input images has been extracted by the FCNN method into the resultant fused image. The MI metric value of the DRPL method is competitive with the proposed method. The MI value of the NSWT method is lowest in comparison to the other MFIF methods. Thus, the fused output image has correlated better as more information is transferred from both far and near focused images on the fused resultant images.

The EI values produced by the FCNN method are higher than the other MFIF methods. High EI value has depicted that the proposed method has given accurate edges differentiating the focused and unfocused regions. This indicated that the fused image had preserved better boundaries than other methods.

The metric SS value for the FCNN method has outperformed other existing MFIF methods. This indicated that the image given by the FCNN method contained the essential characteristics in the image. It has been observed that the HP metric value given by the FCNN method is best in comparison to the existing MFIF methods. This has been noted that the proposed FCNN method has acquired most of the features in the human visual system. The greater MI, EI, SS and HP values have indicated that the proposed method outperformed other existing MFIF methods.

3.5. Ablation experiments

In this section, detailed verification of the effectiveness of the contribution of key components is illustrated. Ablation experiments are conducted using both reference and non-reference metrics on the following aspects.

3.5.1. Impact of proposed model architecture

The proposed model architecture ablation, i.e., contributions of fuzzy sets for the MFIF, is discussed. To observe the influence of fuzzy sets, both qualitative and quantitative evaluation is done with CNN [22]. Images in Fig. 13 from dataset 3 display three sets of multi-focus images employed to verify the FCNN method. Fig. 13(a) represents the reference image, Fig. 13(b) and Fig. 13(c) illustrate the left and right focused input images, respectively.

Fused output results of CNN and FCNN methods are depicted in Fig. 14(a) and Fig. 14(b), respectively. For image 1, it is analyzed that the CNN output image contains distorted pixels. The area shown with blue and yellow rectangles has an unclear and unsharpened object of interest. CNN is not able to fuse the pixels from the left and right parts of

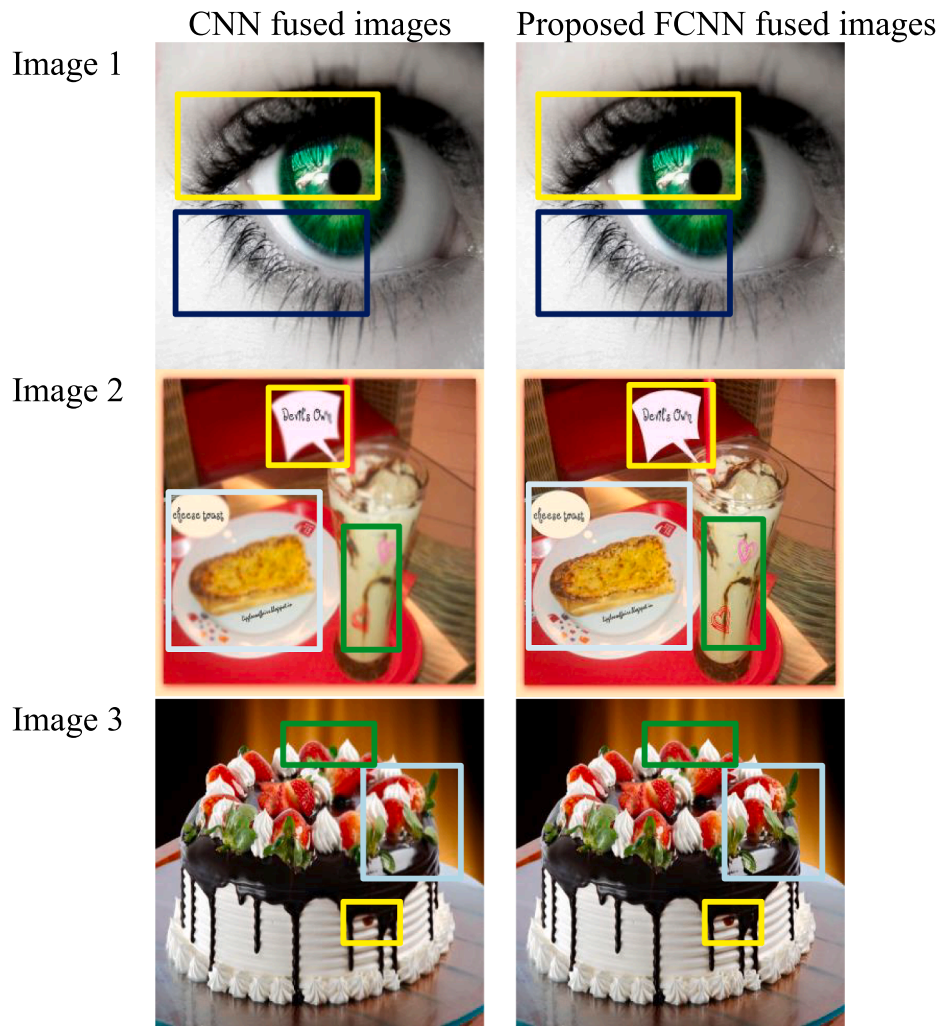


Fig. 14. (a) CNN fused images. (b) Proposed FCNN fused images.

Table 4

Performance evaluation of the 3 images for CNN and FCNN methods using both reference and non-reference metrics.

	Methods	MI	EI	SS	HP	PSNR	RMSE
Image 1	CNN	1.16	0.80	0.90	0.82	36.55	8.01
	FCNN	1.33	0.83	0.93	0.84	38.25	8.39
Image 2	CNN	0.89	0.68	0.84	0.69	39.25	8.11
	FCNN	1.52	0.88	0.99	0.88	55.40	1.64
Image 3	CNN	1.00	0.81	0.92	0.82	40.89	6.78
	FCNN	1.40	0.84	0.97	0.86	45.50	3.84

input images. On the contrary, FCNN has produced sharpened images with better-focused regions shown with yellow and blue rectangles. The FCNN fused images are similar to the reference images displayed in Fig. 13(a).

For image 2, it is clearly shown with three different colored rectangles that the CNN produced image is contaminated with noisy pixels. Hence, as a result, the fused image is blurry and poor in quality. On the other hand, FCNN has generated a full focused and clear visual effect image. The focused regions from Fig. 13(b) and Fig. 13 (c) are merged in the fused image. For Image 3, CNN is not able to generate a sharp and clear image. The fused output image of the FCNN as marked with three rectangles shows that the FCNN produced fused image is better in quality than the CNN image. Hence, it can be summarized that the FCNN model improved the fusion quality compared to the CNN method. Further, dataset 3 is utilized for the quantification of the proposed

Table 5

Average results on Dataset 3 (15 sets) using different types of membership functions.

	MI	EI	SS	HP	PSNR	RMSE
Gaussian	1.388	0.848	0.980	0.878	57.20	1.816
Trapezoidal	1.389	0.852	0.981	0.879	57.21	1.815
Triangular	1.382	0.850	0.977	0.875	56.87	1.824
S type	1.390	0.853	0.981	0.880	57.23	1.814

method. For this purpose, two reference metrics (PSNR and MSE) and four non-reference metrics such as MI, EI, SS and HP are utilized. Results are listed in Table 4. The computed values on dataset 3 using different membership functions with the highest performance is highlighted in bold (see Table 5).

3.5.2. Impact of S-Type membership function

There are many linear and non-linear membership functions that exist. The effect of different membership functions is tested. Comparisons of the metric values indicate the better efficiency of the S-type membership function in achieving better-measured values than other membership functions. Therefore, it concludes that the performance of S type membership degree is better than the linear membership degree.

4. Conclusions

This study proposed a novel integrated pixel-based FCNN method. The presented FCNN method has combined the benefits of FS and CNN to give high-quality fused images. The essential virtue of the proposed integration is that it holistically processed the colored image. Besides, FS is used to automatically remove the uncertainties like blurredness and distortion presented in an image. Thus, it helped to reduce the traditional methods of manually designing the fusion rules and feature extraction. The integration of the FS and CNN has proved to be beneficial in the photography domain as the comprehensive image is produced with less bandwidth and memory. Fuzzy set attempts to solve this problem by considering uncertainty. Subsequently, acquired results are utilized as input to the CNN framework.

Different evaluation metrics have been employed to test the effectiveness of the FCNN architecture that significantly outperformed the other conventional MFIF methods. Thus, experimental analysis has shown that the FCNN method has attained more productive outcome results in quantitative and qualitative measurement. The proposed FCNN method has given the best fusion performance in terms of six assessment measures over 56 images, such as MI is 1.1676, EI is 0.7281, SS is 0.9350 and HP is 0.8020. It presented the open research issues that need to be explored in future work. We could extend this research to other domains of image fusion like multi-modal and multi-exposure image fusion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by Taif University Researchers Supporting Project Number (TURSP-2020/114), Taif University, Taif, Saudi Arabia.

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